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CIRCULAR ECONOMY FOR GLOBAL WATER SECURITY

Water treatment and artificial intelligence techniques: a systematic literature review research

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Abstract

As clean water can be considered among the essentials of human life, there is always a requirement to seek its foremost and high quality. Water primarily becomes polluted due to organic as well as inorganic pollutants, including nutrients, heavy metals, and constant contamination with organic materials. Predicting the quality of water accurately is essential for its better management along with controlling pollution. With stricter laws regarding water treatment to remove organic and biologic materials along with different pollutants, looking for novel technologic procedures will be necessary for improved control of the treatment processes by water utilities. Linear regression-based models with relative simplicity considering water prediction have been typically used as available statistical models. Nevertheless, in a majority of real problems, particularly those associated with modeling of water quality, non-linear patterns will be observed, requiring non-linear models to address them. Thus, artificial intelligence (AI) can be a good candidate in modeling and optimizing the elimination of pollutants from water in empirical settings with the ability to generate ideal operational variables, due to its recent considerable advancements. Management and operation of water treatment procedures are supported technically by these technologies, leading to higher efficiency compared to sole dependence on human operations. Thus, establishing predictive models for water quality and subsequently, more efficient management of water resources would be critically important, serving as a strong tool. A systematic review methodology has been employed in the present work to investigate the previous studies over the time interval of 2010–2020, while analyzing and synthesizing the literature, particularly regarding AI application in water treatment. A total number of 92 articles had addressed the topic under study using AI. Based on the conclusions, the application of AI can obviously facilitate operations, process automation, and management of water resources in significantly volatile contexts.

Keywords Artificial intelligence · Water treatment · Literature review · Water quality

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Introduction

One of the increasing concerns of the developing nations has been water quality. Given the release of polluted effluents into fresh water systems due to the increase of human activities along with industrial movement, accessibility of clean water has become a challenge for many people (Fan et al. 2018). Although around three-quarters of the overall surface of the Earth is covered by water, approximately 99% of it is salty, brackish or frozen, leaving around 1% as drinkable water (Al Aani et al. 2019). Since the global population is increasing with subsequent increase of the climatic burden of pollutants as well as changing the use of land into fresh water ecosystems, availability of clean water is considered of vital importance across the world (Sagan et al. 2020). The polluted water needs treatment before its consumption by the population. In

the same way, plants and animals may be exposed to different hazards of polluted water in the case that their metabolic processes are affected by its consumption (Hasan and Muhammad 2020). Water contamination primarily happens because of organic as well as inorganic waste, sediments, radioactive substances, effluents, sewage, and also heavy metals (Dubey et al. 2015). There are several significant challenges in the universal water conditions, which are primarily the outcomes of global population increase along with climatic changes. Such challenges require new innovations to make sure that fresh water resources are available and water pollution is decreased worldwide (Gehrke et al. 2015). Accordingly, timely purification of contaminated water through sewage treatment seems essential so that the emission standards could be met (Yu et al. 2018).

Dramatic improvements have been recently observed in the treatment of water due to advancements in the modeling processes to support the associated systems. Also, attempts have concentrated seriously on the transformation of the waste management industries to become more sustainable and profitable through developed technologies and smart systems. Modeling and optimizing have the highest importance in the environmental process for the improvement of the systems and enhancement of the procedures' efficacy while keeping the costs as low as possible (Shojaeimehr et al. 2014).

Different methodologies such as regression analysis or "time series" analysis have been applied to water treatment. The benefits of these techniques are in their comparative simplicity along with their convenient implementation. Nevertheless, their relative restricted capabilities of forecasting in specific conditions should be taken into account, particularly when nonlinear associations and high levels of noisy data are present (Adamowski and Karapataki 2010; Rajaee et al. 2019). Thus, with the aim of solving these restrictions, the overall structure of the business will be replaced by the incredibly significant advancements in artificial intelligence (AI) technologies, while the nature of work will be also modified and, correspondingly, extreme changes can be witnessed in people's lifestyles. The recent growth of AI has led to considerable advancements in applications such as autonomous driving, big data, pattern recognition, intelligent search, image understanding, automatic programming, robotics, and human–computer game; therefore, human societies will be affected by this technology to a great extent (Fan et al. 2018; Malviyaa and Jaspal 2021).

AI approach is considered an efficient option for modeling of sophisticated nonlinear systems and identifying complicated associations of input and output data (Ahmed et al. 2019). The application of AI has been reported by several researchers since it is easy to use, while the operations are carried out at high speeds, and its accuracy is acceptable with no requirement to figure out physical issues (Rajaee et al. 2019). The application of AI-based methods to different engineering fields, such as water treatment and desalination, has been also reported due to their capabilities of dealing with practical challenges. These AI methods can significantly reduce the cost of water supply and sanitation systems while ensuring compliance with drinking water quality and wastewater treatment (Hmoud Al-Adhaileh and Waselallah Alsaade 2021). Moreover, AI has been also popular in modeling the plants used for wastewater treatment. In domestic and industrial applications, water quality depends on the control of main variables by achieving the desired impacts of operations and making sure the required standards and protocols have been met (Baklouti et al. 2018).

According to Li et al. (Li et al. 2021), the AI technology presents technical support for the management and operation of drinking water treatment processes, making them more efficient than depending entirely on human labor. They emphasized that water quality diagnosis, autonomous decision making, and operation process optimization are all possible with AI-based data analysis and evolutionary learning processes, and these have the ability to create a universal process analysis and prediction model platform. Based on Russell and Norvig (2002), AI can be considered the investigation and framework of intelligent factors, where an intelligent factor carries out the actions which enhance the potential of success.

AI incudes different branches, among which Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Monte Carlo simulation (MCS), Simulated Annealing (SA), Artificial Neural Networks (ANNs), and hybrid models can be mentioned, along with two or more prior branches with the ability to deal with ill-defined problems, configuration of sophisticated mapping, and prediction of the results (Yetilmezsoy et al. 2011). The efficient and accurate utilization of intelligent strategies lead to the advancement of extensive and beneficial systems which perform better or various features, whose achievement would not be possible when traditional approaches are used (Bhandari et al. 2015). Nevertheless, applying AI tools is still restricted in the field of environmental protection in general, because of their concentration in water treatment (Mandal et al. 2015).

Thus, the present work carried out a systematic collection, summarizing, analysis, and synthesis of the information associated with the practical applications of AI technologies in the treatment of water. As far as the authors know, the study can be considered the first with a focus on a systematic review of literature on the application of these technologies in water treatment. To this aim, a Systematic Literature Review (SLR) approach was used to investigate the possibilities of AI application in water treatment. Accordingly, three questions were raised to attain the research objectives and implement a review of previous studies.

RQ1: What are the initial research works associated with the AI methodologies for water treatment?

RQ2: What are the different AI applications employed for water treatment between 2010 and 2020?
RQ3: What are the strengths, drawbacks, as well as robustness of AI applications in water treatment?

Research on water treatment shows a significant increase during the time. Some contributions regarding the current study are as follows:

i. A total number of 92 initial research works were found in the field of AI technologies and their developed applications to figure out the AI field better during 2010–2020; nonetheless, incorporation of the findings of prior studies along with the assessment of the present state of the research in this area has been almost neglected.
ii. Evaluation of the studies performed in this field can suggest novel paths for researchers in relevant fields such as AI and soft computing with promising ideas to deal with the existing gaps.
iii. A summary of the strengths and restrictions of every model was provided in the present work, which is necessary for future investigations on different water treatment fields, challenges regarding data collection, and recommendations for further improvement.

This paper has been presented based on the following organization. Section 2 presents the methodology employed for the study. Section 3 elaborates the obtained data and a review of the answers to the research questions. Eventually, the last section provides discussions, an in-depth elaboration of the opportunities for, and advantages from more research in the area.

Review method

The SLR method was used to provide the answers to the study questions. Accordingly, three-staged guidelines suggested by Kitchenham and Charters (2007), including planning, conducting, as well as reporting were used. The research questions were identified in the first phase along with the development of the review protocols. The search keywords along with the inclusion and exclusion criteria were determined in the next phase while performing the quality assessments. Finally, data extraction and synthesis took place, after which the findings were reported. SLR method is dependent on precise and assessed protocols for the extraction, analysis, and documentation of the results according to Fig. 1.

Search technique

According to Fig. 1, the search was performed both automatically and manually. Several related papers were obtained using the search terms. Selection of electronic databases consisting of Science Direct, ISI web of knowledge, IEEE explore, ACM Digital Library, as well as Scopus, was performed as the research primary sources, while the databases were expected to cover the related papers. The primary keywords employed included "Water treatment" OR "Wastewater treatment" OR "River water quality" OR "Drinking water treatment" AND "Artificial intelligence" OR "Machine learning". Moreover, a search was performed manually over the reference selection of papers to look for papers not included in the search of the databases.

Inclusion as well as exclusion criteria

All the papers (1) published from 2010 to 2020 (2) from peer-reviewed journals since due to reporting validated findings and having the highest effects on the field (Khosravi et al. 2019)(3) with providing direct or indirect responses to the specified research questions were entered into the study. On the other hand, all papers (1) unrelated to the research questions of the present work, (2) with a non-English manuscript, (3) representing duplications, and (4) not completed were excluded from the study.

The process of study selection

Basically, the main objective of the study selection process has been identifying the related papers for the review of the literature. According to the search criteria, 331 primary papers were obtained in the present work. Following the elimination of the duplicated papers with the use of Mendeley, a total number of 227 papers were left. In the following step, the abstracts and the conclusions of the articles were considered based on the inclusion as well as exclusion criteria. Accordingly, 89 articles were further removed. The suggestions made by Kitchenham and Charters (2007) were followed at this step and all the papers irrelevant to the study were excluded. Full-text scanning with regard to the exclusion criteria was applied for the rest of the papers. In addition, the manual step was also performed for the reference part of the papers to find out if any papers had been missed. Accordingly, 12 further papers could be found by this method. Finally, the application of the quality assessment criteria helped in the elimination of 9 articles, and eventually, 92 papers remained as the final set of primary reports. Appendix 1 indicates the list of the included papers.

Criteria of quality assessment

The Quality Assessment (QA) criteria were employed for all the papers based on the recommendations of Niknejad et al. (2021) in order to assess the quality of

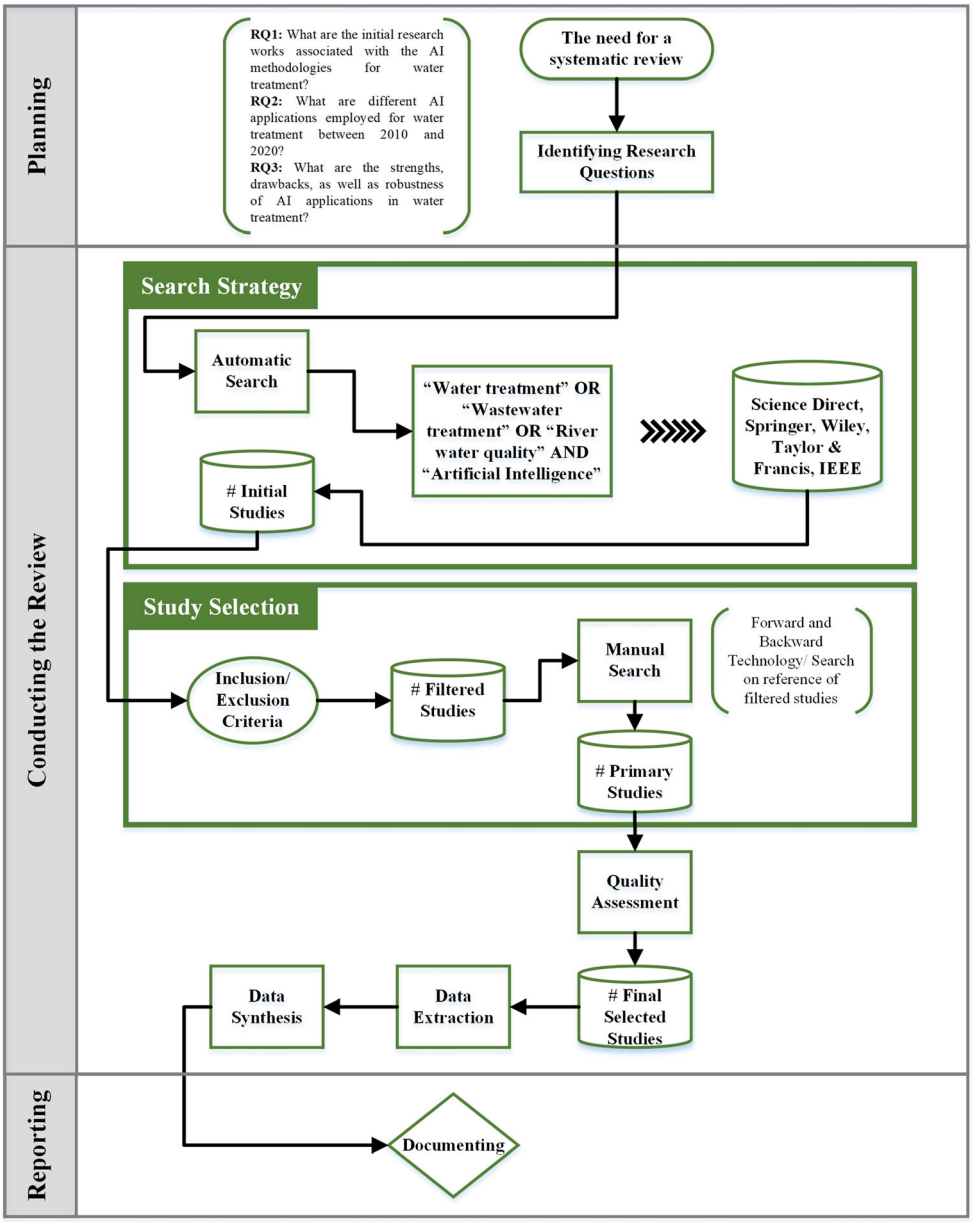


Fig. 1 Review protocol

the papers extracted. The following research questions were used for the QA of the articles:

QA1. Is the topic of the extracted papers relevant to the assessment or monitoring of water quality?
QA2. Is the AI method explicitly mentioned in the article?
QA3. Are these suitable classifications of the data collection strategies or case studies in the papers?
QA4. Is the step of data analysis explicitly described in the articles?

Examination of the QA criteria mentioned above for 92 research papers aimed at determining their reliability. Accordingly, recommendations made by Nidhra et al. (2013) were considered to apply QA, including three quality schema steps of high, medium as well as low, based on which the articles' quality was dependent on their loading scores. Scores over or equal to 5 meant high quality, while scores of 4 indicated the medium and scores of below 4 represented low qualities. Following such a process, 9 papers did not satisfy the criteria and were removed. Totally, according to the QA criteria proposed by Nidhra et al. (2013), the initial papers of

this SLR included 92 articles. Figure 2 displays the distribution of articles based on QA.

Extraction and synthesis of data

The design of data extraction aimed to record the precision of the information obtained by the researchers from the initial investigations. For this purpose, every study was reviewed exactly, and the relevant data were extracted with the use of Mendeley as well as Microsoft Excel spreadsheets. The columns taken into account in the review were as follows: paper ID, to specify every paper, the authors, the publication year, the number of citations for every paper, AI categories, AI applications, case study region publishing location, applications, training datasets, testing datasets, and validation datasets. The selection of the abovementioned items took place in agreement with the study questions and objectives. In particular, the data for the 92 articles were extracted using a form indicated in Table 1.

Results

The publications' temporal aspect

Figure 3 indicates the distribution of 92 papers associated with water treatment from the temporal perspective. According to what is observed, the main published papers applying the AI method to water treatment have been in 2019 with 23 times, while the published papers showed an average of 236 citations. According to Fig. 3, the publications of water treatment research works had a gradual increase from 2014 to 2020 with an average of 68 published papers. Therefore, it can be concluded that attention to this issue has been increased, primarily from 2014 onward. Totally, the citation rate of the papers was relatively acceptable that could indicate their effects and highly considerable quality. According to these statistics, researchers have shown more interest in examining water treatment and AI during the past few years.

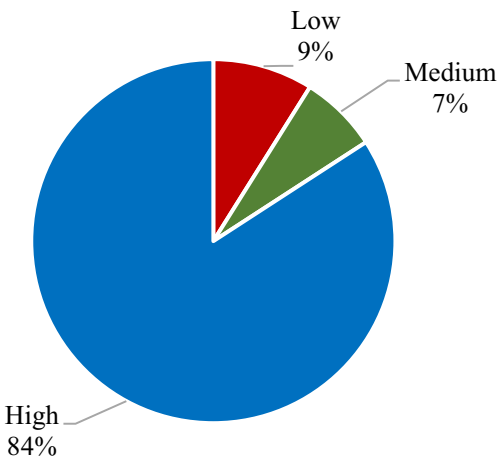


Fig. 2 Distribution of articles based on quality assessment

Figure 4 indicates the distribution of papers according to the cased study region and country of study location. Totally, China had the highest contribution with the maximum number of papers associated with water treatment, wastewater treatment, prediction of river water quality, and drinking water treatment, forming a total number of 13 articles. Iran, India, and the USA allocated the next ranks with 11, 9, and 8 papers, correspondingly. Nevertheless, the case study region of eleven papers was not clear.

Initial studies associated with the AI methodologies for water treatment

Currently, studies associated with AI are following a gradually increasing trend, with the development of different AI algorithms along with their application in various areas. There is a continuous growth in the number of papers discussing around keywords such as "artificial intelligence" and "water treatment" across the world. A search in the Google Scholar database with the use of the mentioned keywords showed the increase of studies performed on AI to predict, automate, control, and improve water treatment performance. Figure 5 illustrates the obtained search results. In the current paper, 92 papers could be extracted from the SLR approach as an initial study. Based on the results, designing and application of different models during the past few years have concentrated on treating water according to AI methods. Appendix 2 presents a summary of the chosen initial studies. Furthermore, some details associated with the extracted papers can be observed as well. The table includes the AI categorizations according to single and hybrid methods, case study region and the application, and applied training, testing, and validation dataset in prior research on the AI as well as machine learning for nonlinear relationship analyses and processes control as well as monitoring in the treatment of water, water quality monitoring, treatment of drinking water, treatment of wastewater, and river water quality.

Different AI applications employed for water treatment

Figures 6 and 7 indicate AI applications in the treatment of water treatment. As it can be observed, categorization of AI methods has been performed on a single and a hybrid method. ANN approach has been the highest frequently used method employed in prior research works. This approach can be regarded as an AI tool typically used to predict the removal of contaminants in different treatment processes because of their potential to self-adapt and self-learn(Al Aani et al. 2019). With access to suitable training algorithms and the correct amount of data, ANNs are capable of solving

Table 1 Data extraction for each study

Extracted data	Description
Study-ID	The unique number of each article
Authors	Names of the authors in the article
Citation	The number of times your article has been cited
Artificial intelligence categories	The category is divided into individual and hybrid method
Artificial intelligence application	Artificial intelligence has various applications for instance: fuzzy logic, deep learning, artificial neural network, etc.
Case study region	Region and country case studies
Application	Water treatment, wastewater treatment, drinking water treatment, and river water treatment.
Training, testing, and validation data sets	The "training" data set is the general term for the samples used to create the model, while the "test" or "validation" data set is used to qualify performance.

multivariate nonlinear problems (Jiao Wang and Deng 2016). They have been often employed in the experimental designs as well for the removal of pollutants over the water/wastewater treatment (Fan et al. 2018). The single and hybrid methods mentioned above will be briefly explained in the next section.

Single methods

Deep learning

Different benefits can be mentioned for the application of deep learning to map water quality (Peterson et al. 2020). These architectures, such as recurrent and convolutional neural networks, seem especially interesting because of significant achievements in current research works and their better performance compared to other methods in different remote sensing applications (Pahlevan et al. 2020). As an example, Sagan et al. (2020) carried out the analysis of current developments in water quality remote sensing and determined the restrictions of available systems, estimation techniques, and suggested future advancements. Based on their conclusions, empirical-based machine learning along with deep learning procedures had a better performance compared to other methods such as band ratios, spectral shape, and bio-optical inversion procedures which had frequent applications.

Mamandipoor et al. (2020) examined a method according to deep neural networks (particularly, long short-term memory) in comparison to statistical and conventional machine learning techniques. Based on their results, significant potentials were found in the application of deep neural networks to manage the faults occurring in wastewater treatment plants, while the suggested technique not only had a better performance compared to the conventional procedures but also the performance obtained a fault detection (recall) of more than 92%, enabling a new class of wastewater treatment plant monitoring and management with less requirement to human monitoring.

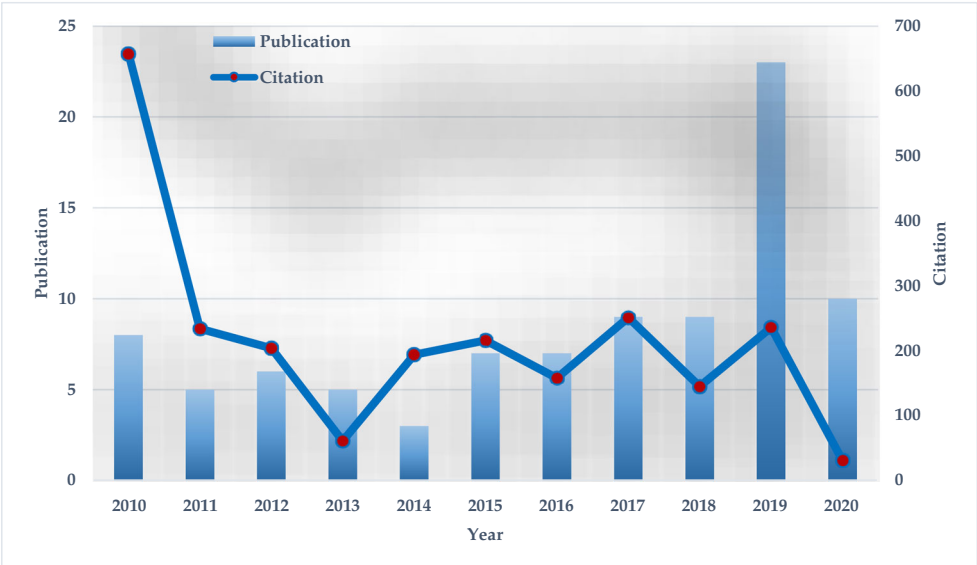


Fig. 3 Temporal view of primary studies during 2010–2020

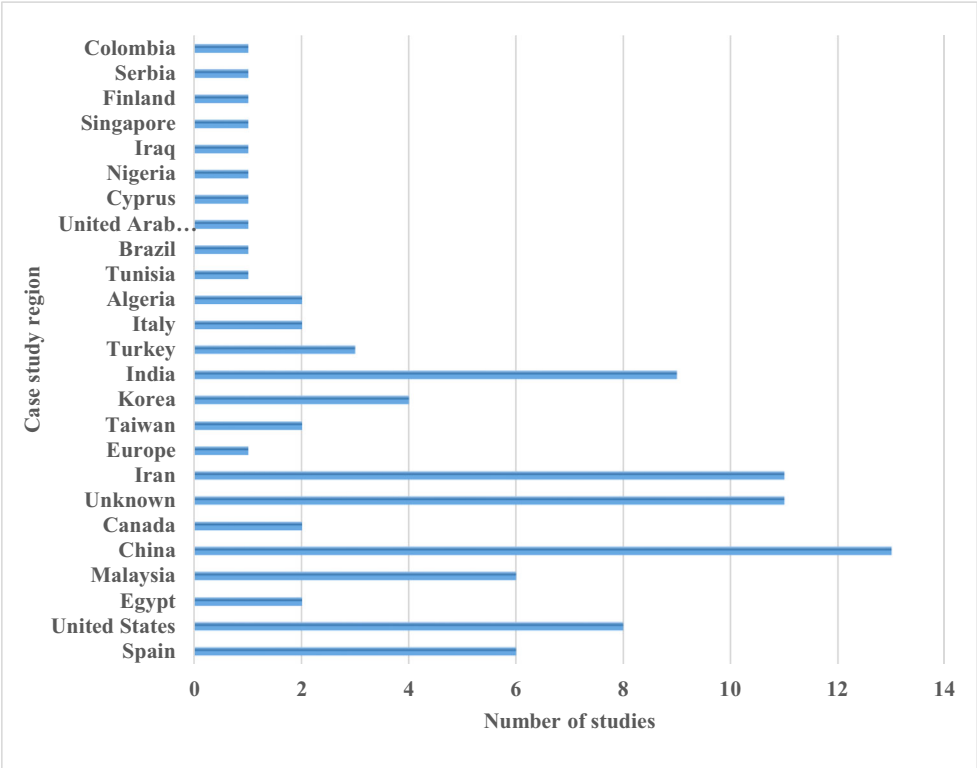


Fig. 4 Distribution of articles based on the country

Artificial Neural Networks (ANNs)

The architecture of ANNs includes three layers of the input layer, the hidden layer which is typically made up of one layer but may include more sub-layers, and finally the output layer (Khalil et al. 2011). Each layer possesses some nodes which have the responsibility of the information transmission from a layer to the following one; however, lateral connections within the layers or feedback connections are impossible (Shu and Ouarda 2007). The ANNs have application in the nonlinear

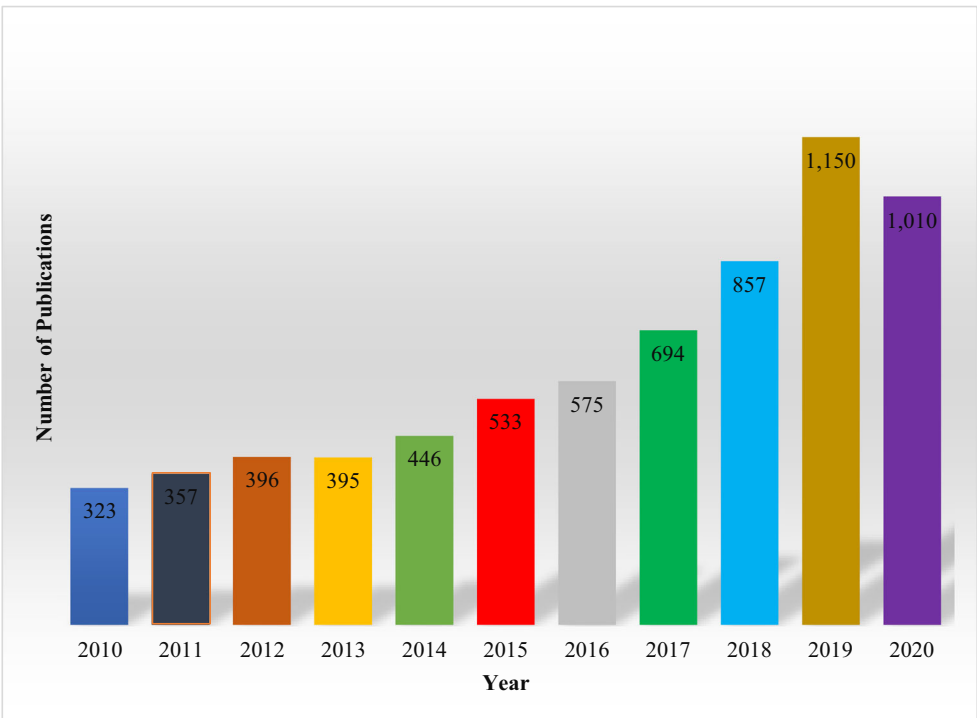


Fig. 5 The annual number of scientific publications on AI and water treatment obtained from Google Scholar

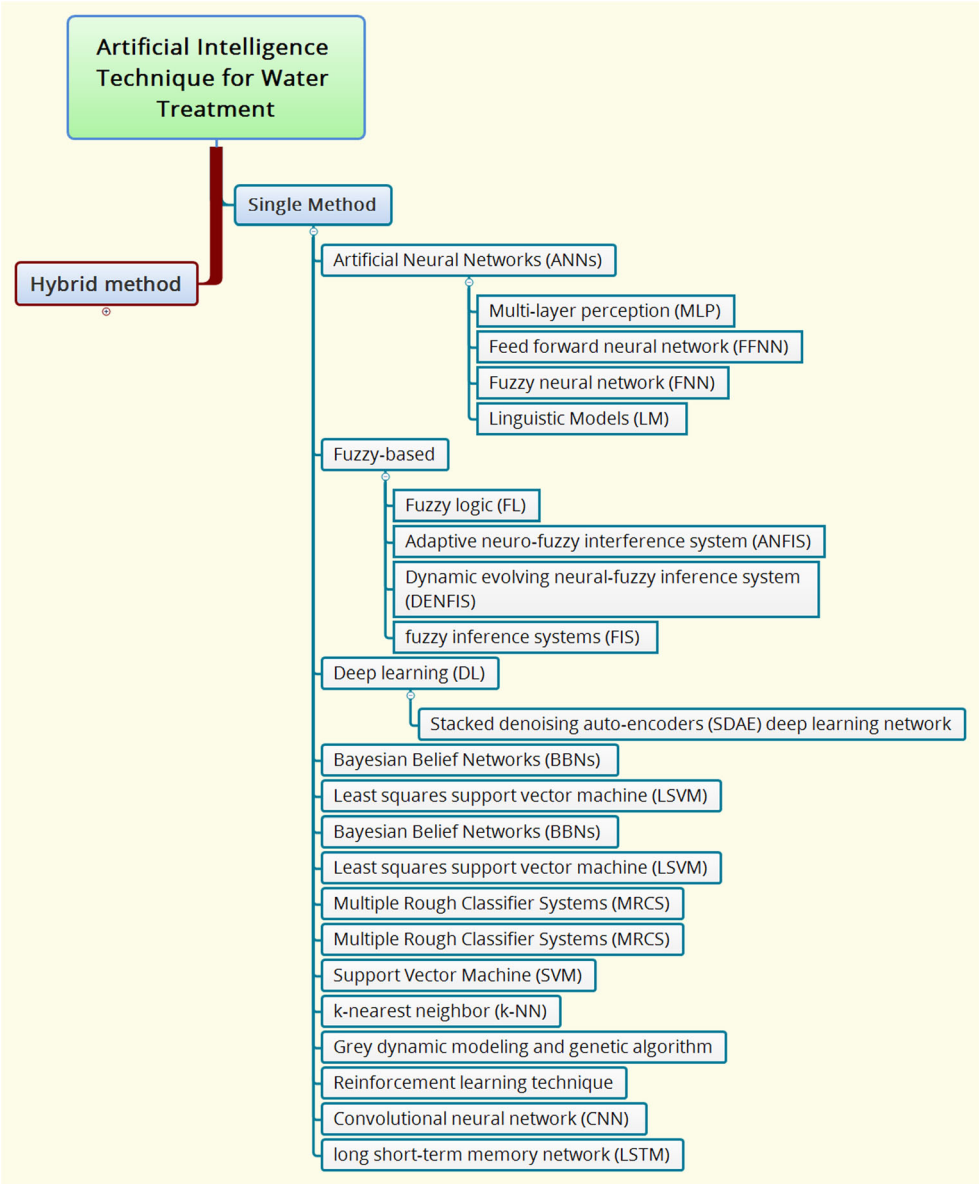


Fig. 6 AI techniques for water treatment (single method)

associations among accumulated input and output numerical data for the coagulation procedures in water treatment (Wu and Lo 2010). Significant attention has been paid to ANNs over the past few years for prediction as well as pattern recognition.

Cabrera et al. (2017) investigated ANNs as a control system tool in Small-Scale Prototype (SWRO) desalination. Their study indicated that the ANNs were capable of managing the operating set-points of the SWRO desalination plants along with the successful adaptation of the plant energy consumption to the extensive and random variations of the existing electrical power. García-Alba et al. (2019) proposed a methodology for the development of ANNs in estuaries as emulators of process-based models for the analysis of bathing water quality along with variability observed during the time and space. Based on their findings, the process-based model and the neural networks could easily reproduce the measurements of *Escherichia coli* concentrations, representing moderately better fitting of the process-based model (R^2=0.87) compared to the neural networks (R^2=0.83).

It is possible to use GA for optimization of ANNs, after which they exhibit desirable predictive robustness during the empirical implementations, representing their capabilities in the behavioral prediction of complicated water treatment procedures (Fan et al. 2018). As an example, Y. Zhang et al. (2019) carried out the development of a prediction instrument to estimate the monthly water production of drinking water treatment plants using a Hybrid Artificial Neural Network (HANN) model developed through GA. According to their findings, the HANN model provided a combination of ANNs and GA benefits, indicating better abilities and strength to extrapolate compared with the single ANN models.



Fig. 7 AI techniques for water treatment (hybrid method)

Honggui et al. (2014) suggested an efficient procedure to detect faults in the Sludge Volume Index (SVI) sensor after which examinations were carried out on an empirical hardware setup in wastewater treatment processes. The design of the SVI prediction plant was provided with the use of a Fuzzy Neural Network (FNN), whose training took place using a historical set of data gathered over fault-free operations of wastewater treatment processes. According to the obtained findings, the suggested fault detection technique was capable of obtaining the fault signals of the SVI sensor online.

Han et al. (2018) developed a multi-purpose optimum control technique, according to the adaptive kernel function, the enhanced multi-purpose particle swarm optimization as well as the adaptive fuzzy neural network controller to identify the optimum control for wastewater treatment processes. Based on the experiments, the suggested multi-purpose optimum control technique could achieve optimum performance and desirable control abilities. According to simulations, the multi-purpose optimum control technique was found to be efficient. Furthermore, Wu and Lo (2010) revealed that the ANN abilities to predict were enhanced through developing the inherent factor with no requirement to normalize data.

Juahir et al. (2010) investigated the quality of Langat River water using several chemo-metrics procedures, including cluster and discriminant analyses along with an ANNs. The ANNs indicated more acceptable prediction performance in

distinguishing the regional area with a considerable favorable percentage of correct classifications in comparison with discriminant analyses.

Turki et al. (2020) studied wastewater treatment in Tunisia through the application of ANNs which indicated the potential of identifying the associations of input and output parameters along with predicting the wastewater treatment process performance with considerable accuracy.

For recovering the water quality of a river, Y. Zhao et al. (2016) designed ANN models for improving the purifying ability of wastewater in wastewater treatment plants of Harbin, northeast of China. The prediction error analyses showed the robustness of the ANN model in simulating and predicting the effluent water quality of the wastewater treatment plants.

Yurtsever and Yurtsever (2019) investigated a novel procedure to classify micro-beads according to microscopic images. Accordingly, they used Convolutional Neural Network (CNN) as a Deep Learning algorithm. The investigation showed the method's capabilities in identifying micro-beads in various cosmetic brands with the use of a CNN. The study was carried out with the CNN, achieving 89% classification performance considering micro-beads in wastewater.

Fuzzy based

Adaptive Neuro Fuzzy Inference System (ANFIS) is supposed to provide more accurate day-to-day estimations of evaporation, predictions of reservoir water levels, and predictions of the river flow (Chang and Chang 2006; Firat and Güngör 2007; Kişi 2006). This technique can be considered a robust fuzzy logic neural network, providing a strategy for fuzzy modeling to obtain information on the datasets which best permit the related fuzzy inference system to trace the desired input/output data (Heddam et al. 2012).

Altunkaynak and Chellam (2010) employed fuzzy logic models in time-variant special fluxes over cross-flow microfiltration of different feed suspensions exposed to a broad scope of hydrodynamic parameters whose derivation and validation were performed. According to their conclusions, a single powerful fuzzy model could carry out a quantitative examination of the cumulative effects of different particle sizes on membrane fouling.

Ahmed et al. (2019) suggested applying improved Wavelet Denoising Techniques with the use of Neuro-Fuzzy Inference Systems (WDT-ANFIS) based on historical water-quality parametric data. According to their conclusions, ANFIS outperformed the alternative MLP-ANN and Radial Basis Function Network-ANN techniques in determining solutions, while it also showed the highest precision and reliability in processing large amounts of non-linear and non-parametric data.

Najah et al. (2012) suggested an enhanced wavelet denoising method with Neuro-Fuzzy Inference System according to the data fusion module to predict water quality. Based on their findings, the optimal accuracy levels were obtained when the length of cross-validation became equal to one-fifth of the data records. Furthermore, the WDT-ANFIS module has a better performance than the ANFIS module while considerable improvements were also observed in the accuracy of predictions.

Heddam and Dechemi (2015) introduced a novel artificial intelligence model named Dynamic Evolving Neural-fuzzy Inference System (DENFIS) according to an evolving clustering strategy to model coagulant dosage rate employed in the coagulation stage. Based on their findings, the trained model performed considerable satisfactorily.

Godo-Pla et al. (2020) employed Fuzzy Inference Systems (FISs) in controlling the combined oxidant dose in the initial disinfection stage of drinking water treatment plants. Their method performed satisfactorily regarding the formation of disinfection by-products based on the utility requirements. The FISs reasoning that can be understood by the users has enhanced their involvement in the application of developed control systems to manage the sophisticated drinking water procedures.

Heddam et al. (2012) examined ANFIS application to model the coagulant dosage in drinking water predictions of Boudouaou, Algeria. Development and investigation of the two ANFIS models were performed, including ANFIS-GRID and ANFIS-SUB, with the grid partition and subtractive clustering techniques correspondingly. Therefore, the study supported higher reliability of the ANFIS-SUB model compared with the ANFIS-GRID to model coagulant dosage.

Wan et al. (2011) employed ANFIS for the development of models to predict suspended solids along with removing the chemical oxygen demand of full-scale wastewater treatment plants which treated process wastewaters from a paper mill. Comparisons were performed with the use of ANN as well. According to the findings, the forecast and control operations were performed reasonably using the proposed system. Moreover, the ANFIS modeling procedure has the capability of predicting performance controlling the treatment process in treatment plants.

Nadiri et al. (2018) investigated the Tabriz wastewater treatment plants and examined their being treatable. Accordingly, an ensemble of fuzzy logic (FL) models were suggested as surrogates for the mentioned plants to prevent the simulation of sophisticated physical, chemical, as well as biologic treatment procedures. Based on their conclusions, the supervised committee FL model had the ability to predict the effluent water quality parameters of wastewater treatment plants with higher accuracy compared to other alternative methods under investigation.

Bayesian Belief Networks

Bayesian Belief Networks (BBNs) have been increasingly adopted as instruments for ecologic and water resource modeling (Molina-Navarro et al. 2020). These networks have the ability to integrate uncertainty in the knowledge related to expert models with the use of cause–effect associations indicated as conditional possibilities (Ortega-Argueta 2020). Furthermore, BBNs employ a computer platform that permits graphic design of conceptual maps that is particularly important given the fact that it is possible to translate perceptions into explicit images which make understanding and learning more convenient (Maani and Cavana 2007). As an instance, Panidhapu et al. (2020) employed BBNs for a probabilistic representation of sophisticated variable interactions for real-time modeling of fecal indicator bacteria level surface waters. According to their findings, BBNs could lead to more accurate predictions by 25–54% in comparison with alternative methods such as logistic regression, Naïve Bayes, as well as random forests.

Francis et al. (2014) used BBNs for the construction of a knowledge model for pipe breaks in a water zone. This model was learned from pipe breaks and covariate data from a mid-Atlantic United States drinking water distribution system network. According to their findings, modeling of drinking water distribution system pipe breaks with Bayesian Belief Networks was both useful and practical.

Support vector machine (SVM)

AVM algorithms have wide application in classification problems including detection of diseases (Richhariya and Tanveer 2018) and diagnosis of faults (X. Zhang et al. 2017). Exclusive universally optimum solutions are provided by this method for the optimization problems. The SVM uses the structural risk minimization principles and is considerably generalizable (Cortes and Vapnik 1995). Given the potential of this method regarding adaptive learning and nonlinear approximation, it possesses distinguished benefits in dealing with small samples and nonlinear data (Jianzhou Wang and Hu 2015).

H. Chen et al. (2020) used a multilayer network as the kernel to optimize least squares support vector machine models for the improvement of the model prediction capabilities. Accordingly, machine learning techniques are improved to determine water pollution quantitatively, while it also gives recommendations to solve problems in water safety.

D. Wang (2016) used the SVM technique to make drinking water treatment more adaptive while stabilizing the quality of water under treatment. An SVM model under treatment of a particle swarm optimization algorithm was employed for the establishment of the real-time assessment model of raw water quality. According to empirical findings, the suggested technique was considerably effective in assessing raw water quality and the feed-forward–feedback control schemes.

Dongsheng et al. (2017) proposed SVM-based raw water quality assessments for making drinking water treatments more adaptive while stabilizing the quality of water under treatment. A feed-forward–feedback control scheme according to the raw water quality assessments was provided to control the chemical dosing process of drinking water treatment to adapt the raw water quality variations while stabilizing the quality of water under treatment. As the empirical findings indicated, the performance of the control scheme increased in comparison with the prior Proportion–Integration–Differentiation (PID) feedback schemes.

K-Nearest Neighbor (k-NN)

the k-Nearest Neighbor algorithms belong to the instance-based learning algorithms, estimating the categories of unknown instances, according to labels of their (k) nearest neighbors (Fejfar et al. 2013). The k-NN algorithm can be regarded as an uncomplicated, but robust and flexible, pattern recognition technique. In the case of categorical target variables, the algorithm is employed to classify tasks (Arroyo and Maté 2009). Moreover, k-NN is the basic and most naive classification strategy in the case of little or no previous knowledge on data distribution (Imandoust and Bolandraftar 2013).

Kim et al. (2016) used the k-NN algorithms to predict the influent features of wastewater treatment plants. Totally, it was indicated that the k-NN technique could be conveniently employed for the prediction of the wastewater treatment plant influent flow rates as well as water qualities in dry climates. It was also suggested that this technique had the potential of predicting wastewater treatment plant influent flow rates as well as water qualities.

Hybrid methods

According to Fan et al. (2018), the hybrid methods' performances regarding the treatment of water could typically indicate how precise and robust they have been with the low MSE rates and large R^2 rates compared to a single method of artificial intelligence. As an example, in the study of Najafzadeh and Zeinolabedini (2019), the application of different soft computing procedures such as Feed Forward Back Propagation Neural Network (FFBP-NN), radial basis function neural network (RBF-NN), ANFIS, and SVM aimed at predicting daily flow rates for the wastewater treatment plants. The suggested models had promising applications in a highly vital component of wastewater treatment plants together with complication of treatment procedures.

Torregrossa et al. (2018) employed ANNs and Random Forest (RF) for the generation of high-performing energy cost models associated with wastewater treatment plants.

According to their conclusions, the model performance indicators were totally more desirable in machine learning algorithms compared to the ones indicated in previous studies.

Hussein et al. (2019) suggested a novel machine learning technique which combined the benefits of the Random Vector Functional Link Network (RVFL) with the Moth Search Algorithm (MSA) with the aim of predicting the missing values of overall algal count while monitoring water quality of surface waters which provided drinking water treatment plants in Fayoum, Egypt. The higher performance of MSA-RVFL was reported compared to the SVM and ANFIS models given various sizes of training tests.

Guo et al. (2015) used two machine learning models of ANNs and SVMs for the prediction of 1-day interval overall nitrogen concentrations of effluent from a wastewater treatment plant in Ulsan, Korea. Based on their findings, the application of both models was effective in the study conditions. The predictions of the SVM model were more accurate in the training step while the same results were obtained for the validation step.

Faruk (2010) employed a hybrid technique, which combined seasonal Autoregressive Integrated Moving Average (ARIMA) and neural network back-propagation models for the prediction of water quality time series data. The ARIMA results were poor representations of water quality data patterns for boron a well as dissolved oxygen, while desirable results were obtained for water temperature. The ANNs could provide predictions of water quality parameters at the suggested time step with higher accuracy.

Sharif et al. (2015) focused on determining the spatiotemporal patterns of the water quality data and identification of the pollution sources in the Klang River Basin. The advanced methodologies of Self-Organizing Map (SOM) together with the K-means algorithm were used in their study. According to their findings, instead of employing the traditional procedures of benchmarking schemes, machine learning algorithms provided the most desirable solutions to recognize patterns.

Saha et al. (2017) used the Non-structural Fuzzy Decision Support System (NSFDSS) along with ANN for identification of the parameters while assisting the decision-makers to provide effective operating systems for water treatment plants. Employment of the NSFDSS indicated the effectiveness of the Clariflocculator as the most appropriate parameter for the water treatment plants.

Hernández-del-Olmo et al. (2019) used machine learning methods such as SVM, KNN, Decision Trees, RF as well as Gaussian Naive Bayes to control the wastewater treatment plants. Accordingly, a roughly 85% accurate performance was achieved in the weather soft-sensor with two machine learning algorithms of KNN and Random Forests to deal with sophisticated control functions in water treatment plant processes.

Liu et al. (2013) proposed a hybrid learning technique that combined GA and ANFIS for the estimation of effluent nutrient concentrations in a full-scale biological wastewater treatment plant. The suggested model could be particularly beneficial in considerable nonlinear processes such as wastewater treatment procedures. The hybrid GA-ANFIS in the full-scale plant carried out predictions satisfactorily.

Manu and Thalla (2017) used the AI models of SVM and ANFIS to predict effluent Kjeldahl Nitrogen concentrations obtained from a biological wastewater treatment plant. The first model led to higher reliability of the predictions in comparison with the ANFIS. Of ANFIS models, Gbell MFMODEL showed slight efficiency in modeling the nonlinear time series.

Bagheri et al. (2015) used Multi-Layer Perceptron Neural Networks as well as Radial Basis Function Artificial Neural Networks (MLPANN and RBFANN) to treat wastewater. According to their findings, having low experimental input data values for training ANNs, the MLPANN have higher precision models in comparison with the RBFANN models because of their higher coefficient of determination (R^2) along with lower values of root mean squared errors.

Al Aani et al. (2019) considered the application of artificial neural networks as a regression model and GA as a universal optimization technique to the desalinization and treatment of water for different purposes. Based on their conclusions, the application of AI tools would obviously direct the water sector toward enhanced operations, automation of processes, and management of water resources in a significantly volatile context.

Nourani et al. (2018) employed three various AI-based non-linear models, including Feed Forward Neural Network (FFNN), ANFIS, SVM approaches along with a classical Multi-Linear Regression (MLR) technique to predict the Nicosia wastewater treatment plant performance. According to the comparisons of single models, ANFIS outperformed other single models regarding both calibration and verification phases. Based on the results, SVM showed higher reliability in comparison with the MLR models. Moreover, in the verification step of biological oxygen demand modeling, the performance of models had higher accuracy up to 10%, 15%, and 16% regarding FFNN, SVM, and MLR models, correspondingly. From ensemble methods, neural network ensemble showed higher robustness along with efficiency in combination and led to the performance improvement of AI modeling up to 24%.

Abba et al. (2020) proposed an emerging evolutionary data-intelligence model named Extreme Learning Machine (ELM) in combination with Kernel Principal Component Analysis (KPCA) for the prediction of the Tamburawa water treatment plant performance in Kano, Nigeria. A conventional FFNN along with a classical linear Autoregressive (AR) model was used to provide a comparison of the prediction

performance. According to the prediction results, KPCA-ELM performed a considerable accurately compared with the single models in all the desired variables, while the only exception was reported considering the prediction of pH.

Y. Zhang et al. (2019) introduced HANN as a hybrid statistic model, combining ANNs with GA in order to forecast the total performance of drinking water treatment plants. The scenario analyses indicated the capabilities of the HANN model in the prediction of water production variations according to the parameter variations, representing the HANN model as an acceptable management tool for decision-makers and managers of drinking water treatment plants to carry out their planning based on regulatory changes, source water quality variations as well as market demands.

Aghel et al. (2019) used two data-driven models of ANFIS and a hybrid adaptive neural fuzzy inference system structure trained by PSO strategies for the prediction and simulation of water quality parameters. Based on their findings, both ANFIS and PSO-ANFIS models were capable of accurate prediction of the output. Furthermore, the PSO-ANFIS models had higher accuracy compared to the ANFIS models, due to their less sophisticated structures.

Afshar and Kazemi (2012) used a 2D hydrodynamic model in the simulation of thermal and water surface elevation profiles in the Karkheh reservoir. The suggested model could overcome the high computational efforts needed in the case of applying traditional calibration search strategies, but on the other hand, the quality of the final calibration results was maintained. Estimator ANNs embedded in PSO in calibration processes could obviously reduce implementation time, while the reliability of the quality of the answers was also satisfied.

Lu and Ma (2020) introduced two hybrid decision tree-based models of Extreme Gradient Boosting (XGBoost) and RF, employed for the prediction of six water quality indicators, consisting of the temperature of the water, dissolved oxygen, pH values, specific conductance, turbidity, and fluorescent dissolved organic matter. Based on the analyses, predictions of Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) with RF and CEEMDAN-XGBoost were more stable compared to other benchmark models. However, the models introduced in the paper are previously capable of achieving higher accuracy in their predictions.

Juntunen et al. (2012) employed linear (Multiple Linear Regression) and nonlinear (Multilayer Perceptrons) modeling strategies to provide models of turbidity and residual aluminum during water treatment processes. Based on their conclusions, both methods showed relatively better goodness of the nonlinear compared to the linear model. Furthermore, MLR was better for applications requiring explicit models or faster calculations, such as in the case of adaptive soft sensors.

Heddam et al. (2019) introduced FFNN, Gene Expression Programming (GEP), adaptive neuro-fuzzy inference system with grid partition (ANFIS-GP) and adaptive neuro-fuzzy inference system with subtractive clustering (ANFIS-SC as highly robust AI methods for the prediction of the phycocyanin pigment concentration as a function of different water quality variables. Based on their conclusions, all the models under study were really desirable and could provide acceptable results while ANFIS_SC was highly accurate compared to the other ones.

Lotfi et al. (2019) provided a combination of the new hybrid linear stochastic model (ARIMA) and nonlinear Outlier Robust Extreme Learning Machine Technique (ORELM) with different preprocesses with the aim of modeling the quality parameters of effluent wastewater. Based on their findings, the employment of hybrid models led to an improvement in the capabilities to predict all quality parameters while the highest performance was obtained for the effluent biochemical oxygen demand model ($R^2 = 0.99$).

Mandal et al. (2015) suggested a strategy to remove As (III) from water solution using a new hybrid material. The empirical findings of their study can be applied in the development of a reliable model according to Back Propagation (BP) learning algorithm with artificial neural networking (BP-ANN) to predict the efficacy of removal. The BP-ANN model efficiency can be examined considering the value of the absolute relative percentage error (0.293) as well as the correlation coefficient ($R^2 = 0.975$).

Asadi et al. (2017) carried out a study according to a three-year data collection period from Detroit Water and Sewerage Department. Among the four popular data mining models chosen for the study (Multi-Adaptive Regression Spline (MARS), ANNs, RF, *k*-NN), those provided by MARS yielded the most optimum estimations regarding effluent biochemical oxygen demand, effluent overall suspended solids, effluent overall dissolved phosphorous, as well as effluent overall suspended phosphorous. Moreover, based on the study findings, significant energy saving was possible in addition to maintaining the water quality within the limits.

X. Zhu et al. (2019) investigated the application of machine learning techniques ANNs with RF to predict metal sorption onto biochars, indicating higher accuracy along with predictive performance for adsorption efficacy ($R^2 = 0.973$) compared to the ANN model ($R^2 = 0.948$). The corresponding significance of variables provided correct directions toward the improvement of heavy metal treatment in the actual-world conditions.

The study of Elkiran et al. (2019) aimed primarily at developing single and multi-step ahead prediction models for dissolved oxygen as one of the significant water quality parameters, at three various stations on Yamuna River in India with the use of AI-based ensemble procedures. Accordingly, three single AI-based models were employed consisting of Back Propagation Neural Network (BPNN), ANFIS, SVM as well as a linear ARIMA model along with three various ensemble

procedures of Simple Average Ensemble (SAE), Weighted Average Ensemble (WAE), and Neural Network Ensemble (NNE). Based on their findings, NNE increased the average performance through single models up to 14% in the verification phase in all the stations under study.

Emamgholizadeh et al. (2014) utilized the applications of Multilayer Perceptron (MLP), Radial Basis Function (RBF), and ANFIS models for the prediction of water quality parameters in Karoon River (Iran). According to their findings, a single-layerMLP/BP with 12 neurons, a single-layer RBF with 25 neurons and an ANFIS model with 3 membership functions could be regarded as the most desirable case with the highest potential of being executed. According to the results of the study, the ANN could be employed effectively in computing and predicting the river water quality parameters.

Jin et al. (2019) recently developed an efficient data-driven model to predict the quality of surface water and perform the analysis of the inherent water quality variation tendencies while providing real-time early warnings based on the historical data obtained through observations. Integration of the proposed data-driven model by an improved GA was performed for the selection of optimum primary weight parameters of ANN and a BPNN for adjustment of suitable connection architectures of neural networks. Based on the prediction results, the proposed data-driven model could lead to significant improvements in yielding accurate and reliable predictions, with the effective provision of real-time early warnings for emergency responses.

Li et al. (2019) have suggested the Recurrent Neural Network (RNN) model with better Dempster/Shafer(D-S) combining three RNN models with improved evidence theory to make the model predictions more accurate and stable regarding various water quality parameters. This newly proposed model is considerable accurate and more stable. Eventually, the engineering application of the RNNs-DS algorithm could be achieved on the self-developed water environmental monitoring and forecasting systems, capable of providing efficient support to assess risk as early as possible and perform preventive measures in water context.

The strengths, drawbacks, and robust aspects of applying AI technique in water treatment

Optimizing water treatment techniques has been a significantly desirable method for the plants' sustainable operational performance. The application of appropriate AI methods would help accomplish such an objective since an optimal process can be established from the beginning to end up with the most favorable configuration along with an integrated system. AI and machine learning tools have a significant performance in fitting complicated associations and predictions in the treatment of water. Such strategies can process data with nonlinear associations whose fitting with a single mathematical model would be challenging. Successful application of the AI methods has been reported in various water treatment processes. The AI system is supposed to identify different conditions which are experienced over the process of water treatment and promote the effectiveness of water production so that stable water quality could be achieved. According to Du et al. (2018), the primary benefits of these methods can be mentioned as their low costs and convenient operation, and therefore these technologies are supposed to be applied extensively.

AI unique instruments such as ANNs, as a regression model, GA as a universal optimization technique, FL as a human expert decision making as well as ANFIS, deep learning, hybrid artificial intelligence, BBNs, etc. have shown to be widely applicable to desalinizing and treating water for a variety of objectives. According to Fan et al. (2018), AI technologies are capable of achieving the recovery of clean water, energy, as well as different materials over the process of wastewater treatment. Reusing wastewater can enhance the environmental quality along with generating economic advantages and leading to more water savings. Based on Rajaee et al. (2019), extensive application of AI techniques can be associated with the fact that these methods are simple, accurate, and flexible, bringing about desirable results. Accordingly, a lot of scholars and professionals have examined the performance of these strategies with no requirement of comprehending and dealing with physical challenges. According to Li et al. (2021), the AI methods are more appropriate for providing models of the water treatment processes in real contexts since they are not dependent on any assumptions related to ideal circumstances. These techniques can be established more conveniently and faster, while they concentrate on being predictable instead of only being model fit. In a similar way, Chaves et al. (2004) showed that the use of stochastic and AI methods, including fuzzy set theory or neural networks, while developing decision support systems would result in appropriate ways for the analysis of all such sophisticated connections and uncertainties. Moreover, Mohtar et al. (2019) mentioned that, AI methods can be significantly beneficial in solving complicated issues of nonlinear water quality data. These methods have the required reliability, permitting convenient decisions for policy-makers, while their implementation is also convenient, efficient, and cost-effective, dealing with data at larger scales.

Nevertheless, the employment of AI techniques in water treatment has still several restrictions. As an example, the AI model which is primarily employed in water treatment is the supervised learning model, with considerable dependence on previous information. In comparison to unsupervised learning, the AI model obtains restricted knowledge through mining (Li et al. 2021).

Moreover, as Chowdhury and Sadek (2012) have mentioned, one of the primary criticisms of different AI paradigms

(such as neural networks) is that they are typically considered black boxes which solely aim at mapping the associations of the output and input variables according to a training dataset. Accordingly, several concerns may be raised considering their generalizability to conditions without good presentation in the dataset. One of the other problems and restrictions for machine language is the deviation of the system from the desired results in the case of applying modifications. As an example, application of the genetic programming aimed at developing the model for estimation of the membrane deterioration in the membrane integrity tests. However, a different phenomenon could be reported in the membrane integrity tests for the large membrane breach (Suh et al. 2011).

Discussions, future directions, and conclusions

In the field of water treatment, AI and machine learning have shown to be quite efficient at fitting complex relationships and making predictions. These approaches can process data with nonlinear relationships that are hard to adapt with a single mathematical model. According to Li et al. (Li et al. 2021), AI has been effectively utilized in a variety of water treatment processes, including water quality prediction, coagulant dose prediction, advanced control of membrane fouling, discrimination of the disinfection by-products formation potential, and membrane preparation and optimization. Furthermore, AI technologies offer a novel approach to better understanding of complicated water treatment phenomena and can support decision-making in water treatment management systems.

Despite various benefits of AI, there are significant drawbacks that have hampered the broad use of these technologies in water treatment systems in the real environment. According to Alam et al. (Alam et al. 2022), some of the primary issues that need to be addressed are data availability and selection, poor reproducibility, and the lack of proof of applications in real-world water treatment.

A study conducted by Corominas et al. (Corominas et al. 2018) was focused on the data analysis in decision support to improve the operation of the wastewater treatment methods using the most cited techniques, i.e., principal component analysis (PCA), ANN, clustering, fuzzy logic, partial least squares regression, and independent component analysis. The result of their study showed that although an excess of methods has been developed and tested, there are still issues to be investigated, including the objective comparison of the techniques, the lack of collaborative approaches, the restricted full-scale validation, the absence of guidelines, and the restricted alternatives for active optimisation of data quality.

Moreover, it is undeniable that the most traditional and typically used procedures for the treatment of water have worked according to regression methods. One of the advantages of these methods is their relative simplicity along with convenient implementation. Nevertheless, their relative limitations in their capabilities of forecasting in specific conditions should not be ignored, particularly when nonlinear associations are present together with high levels of noisy data (Adamowski and Karapataki 2010). Thus, to model and optimize the removal of pollutants and water treatment, an AI method is provided as the primary instrument in the experimental design with the ability to generate the optimum operational variables, since to the considerable recent advancements of AI. In this regard, the present work seeks to come up with an overview of water treatment according to AI techniques. Accordingly, the systematic literature review approach has been employed to answer the determined three study questions. Hence, identification of the related studies took place using a keyword search of the works carried out on June 2020, from 2010 to 2020. Following different systematic procedures, a total number of 92 papers could be found with a focus on water treatment and AI procedures. In the current study, two categories of modeling methods, i.e., single and hybrid strategies, are introduced. Within each of these categories, a short illustration of the typically employed techniques for water quality modeling has been presented. According to the analyses performed on the selected papers, there is currently an increasing trend toward the application of AI techniques to water treatment and monitoring of water quality. Out of different AI techniques, ANNs have been known as an appropriate and robust method for such applications, and they have gained more popularity in modeling different water quality variables compared to other AI models. Furthermore, ANNs have the ability to overcome the restrictions of traditional mathematical models through the extraction of the necessary information with the use of training data, with no requirement to a previous specification of proper fitting functions. In addition, based on the results of the literature review, machine learning algorithms propose better solutions for pattern recognition compared to the application of traditional benchmarking schemes. Besides, when the single and hybrid methods were compared, it was found that the performance of the latter in the treatment of water showed its frequent accuracy and robustness compared to the former, which shows agreement with prior research (Fan et al. 2018).

Fan et al. (Fan et al. 2018) stated that investigations should focus on the removal of pollutants such as nutrients, heavy metals, and persistent organic pollutants using AI tools because of their self-learning and self-adapting. ANNs can overcome the limitation of conventional hybrid models using optimization algorithms such as GA or Particle Swarn Optimization (PSO). The authors emphasized that because the weight and bias between neurons are assigned at random, ANNs have poor reproducibility. Therefore, they suggested using the hybrid of ANNs and other AI techniques such as PSO, decision tree, and colony algorithm with the purpose of

determining the best operational factors for the removal processes.

Zhao et al. (L. Zhao et al. 2020) provided a bibliometric analysis on AI focusing on the technology, economy, management, and wastewater reuse. Based on their study, Neural-fuzzy and ANN-GA were the most used hybrid models in the prediction and optimization of the removal of conventional pollutants, typical heavy metals, organic pollutants, and mixed pollutants. Zhao et al. (2020) stated that AI technologies commonly utilize the Neural-fuzzy and ANN-GA models in wastewater treatment management to assist the simulation, evaluation, prediction, and diagnosis of wastewater treatment operations.

Finally, according to the brief comparison of traditional and AI methods, determined AI procedures were effective tools in water treatment. Furthermore, a discussion of the drawbacks of AI techniques was also provided along with its advantages and strengths.

As far as the authors know, the present work can be considered the first study on water treatment, considering different modeling approaches along with numerous variables of water quality. Although a lot of studies have been performed on the application of AI in water treatment, long-term predictions have attracted less attention to themselves. Modeling of different physical, chemical as well as biological variables has not been carried out with the use of these models. Therefore, there is evidently a shortage of extensive studies on the utilization of such approaches in modeling different variables associated with water quality. Consequently, making reliable comparisons and deriving certain conclusions considering different modeling approaches will be relatively challenging. As a result, it is obviously a serious requirement to carry out more studies and develop powerful AI approaches. According to the 92 articles carried out on water treatment, wastewater, and drinking water along with the review of river water quality, it is recommended that combined or hybrid approaches can be taken into account compared with the sole application of AI methods; thus, development of these combined forms should be considered in future studies.

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